Networks, Topologies and Categories

Lecture 01

Computer Networks and Distributed Programming

What this course is about

- We start at the top, with the applications you already use every day.
- Then transport: how a byte stream survives a network that loses things.
- Then the network layer: addresses, routers, and how a path gets chosen.
- Then link and physical: frames, errors, collisions, and actual wires.
- Along the way you write programs in C, on real sockets.
- We close with security and privacy—what happens and what to do when somebody is listening, or lying.

Today: what a network is made of, and how networks are classified.

What a computer network is

Definition

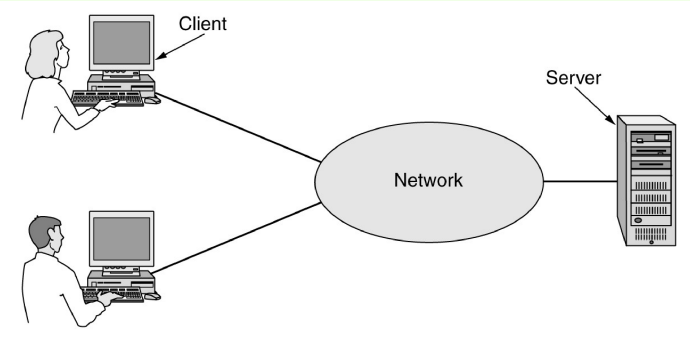
A computer network is a collection of interconnected, autonomous computing devices. Two devices are interconnected if they can exchange information.

- Autonomous matters: no device is in charge of the others.
- The interconnection may be copper, fibre, or radio: the definition does not care.
- Networks are themselves connected into larger networks — the Internet is one.

Uses:: access to information, person-to-person communication, commerce, entertainment, and the Internet of Things.

Access to information: client and server

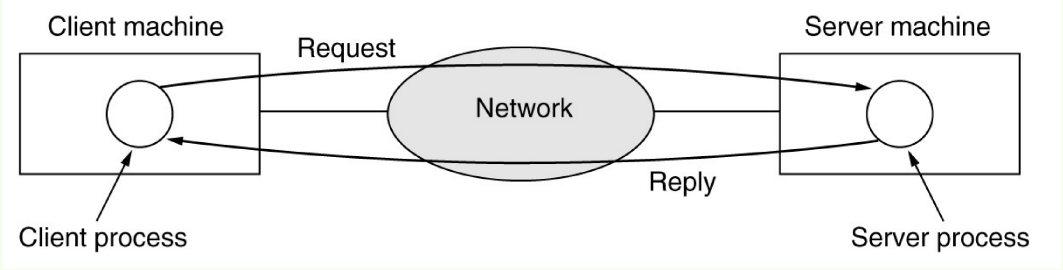
A client explicitly requests information from a server that holds it. One server routinely handles hundreds or thousands of clients.



A network with two clients and one server

Client and server are processes

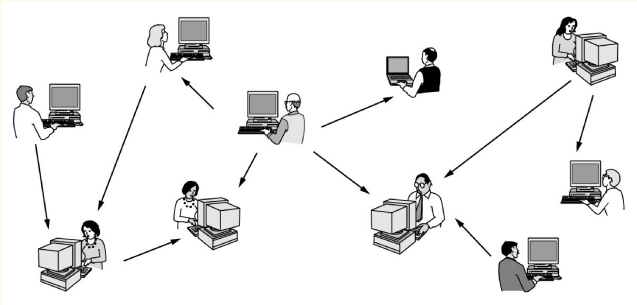
What communicates is not a machine but a process: the client sends a request message and then waits for a reply.



Two processes, one on each side, is the shape of almost everything we will write.

Peer-to-peer: no fixed clients or servers

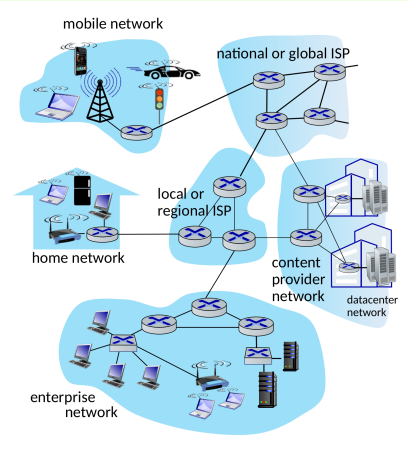
In a peer-to-peer system every participant can act as both sides, and there is no central database of what exists: each peer asks the peers it knows.



In a peer-to-peer system there are no fixed clients and servers

The Internet is a network of networks

Home, enterprise, mobile and data center networks, each managed independently, interconnected through providers into one system that behaves as a single network.



ARPANET, 1969

- Funded by ARPA, a research network for universities and research centres.
- Four nodes by the end of 1969: UCLA, SRI, UC Santa Barbara, Utah.
- Similar networks were being built independently in the US and in Europe.



The initial ARPANET: SRI, UCLA, UCSB and Utah

From a research network to a commercial Internet

- 1974 — Kahn and Cerf design TCP and IP.
- 1986 — NSFNET, a TCP/IP backbone for research.
- 1990 — ARPANET is decommissioned.
- 1995 — NSFNET goes too, and the last restriction on commercial traffic falls with it.



The NSFNET backbone in 1992

Both early networks carried commercial restrictions: the Internet we use begins when they are lifted.

Devices, links and hosts

Infrastructure: network devices and the links between them.



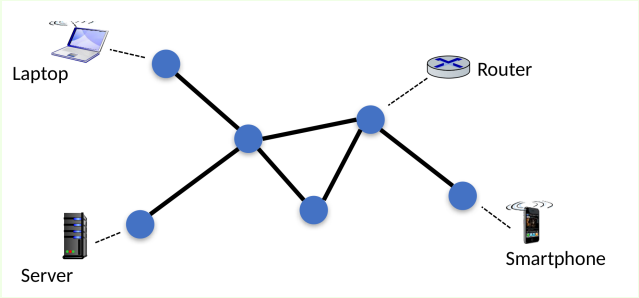
Hosts: the machines that run the applications — also called end systems.



Routers and switches are both packet switches: they receive a packet on one link and forward it on another.

A network is a graph

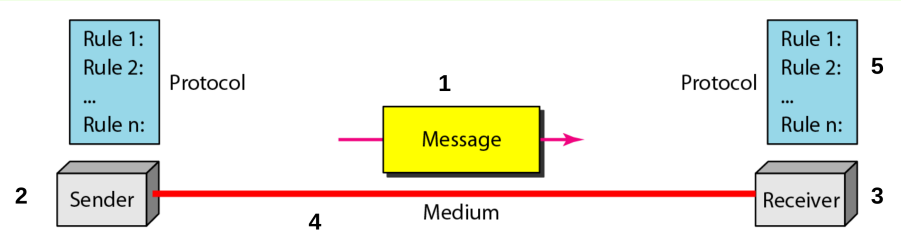
- Devices are nodes, connections between them are links.
- A sequence of nodes and links from one end to the other is a path.
- Hosts are the special nodes that provide or use services.



Ideally hosts are the leaves; intermediate nodes forward

The five components of a data communication

1. Message — the data to be transmitted.
2. Sender — the entity sending it.
3. Receiver — the entity meant to get it.
4. Medium — the channel the message travels through.
5. Protocol — the rules, known to both, that make the exchange work.



What a protocol is

Definition

A protocol defines the format and the order of the messages exchanged between two or more communicating entities, and the actions taken when a message is sent or received.

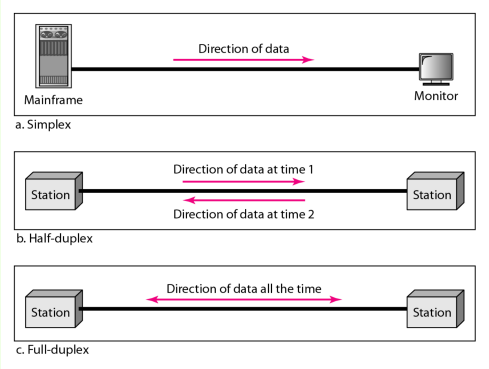
- Asking a stranger the time is a protocol: greeting, greeting back, then the question.
- Fetching a web page is the same shape: connection request, reply, then the request itself.
- Two entities running different protocols get no useful work done.

Everything in this course that involves two remote parties is governed by one.

Data representation and data flow

Data can be text, numbers, images, audio or video: on the wire it is all bits.

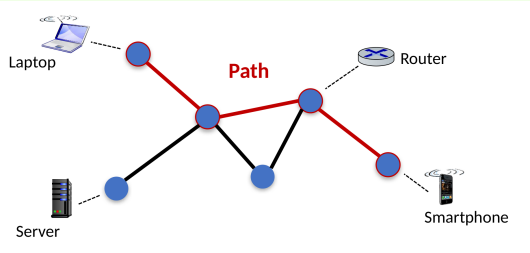
- Simplex — one direction only.
- Half-duplex — both directions, taking turns.
- Full-duplex — both directions at the same time.



Rate, bandwidth and throughput

- Transmission rate — the maximum a device can push out, in bit/s.
- Bandwidth — the maximum a path, links and nodes together, can carry.
- Throughput — what is actually flowing along that path right now.

The first two are ceilings, the third is a measurement, and the third is never above the second. Note that bandwidth here means a rate; when we reach the physical layer the same word will also mean a width in frequency.



A path across the network

Packets and store-and-forward

- A long message is broken into packets, each with a header of its own.
- A packet of L bits on a link of rate R takes L/R seconds to transmit.
- A switch is store-and-forward: it receives the whole packet before sending the first bit onward.
- Across N links of rate R, with nothing else in the way, one packet costs $N\,L/R$.

Consequence

Each output link has a queue. If packets arrive faster than the link drains, they wait; if the queue is full, a packet is dropped. This is where loss comes from, and we will build on it repeatedly.

Circuit switching and packet switching

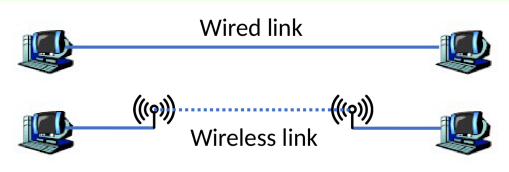
- Circuit switching reserves rate along the whole path before anything is sent, and holds it for the session — the telephone network.
- A circuit is carved out of a link by dividing frequency (FDM) or by dividing time into slots (TDM).
- Packet switching reserves nothing: packets use the link on demand and queue when it is busy.

Why the Internet chose packets

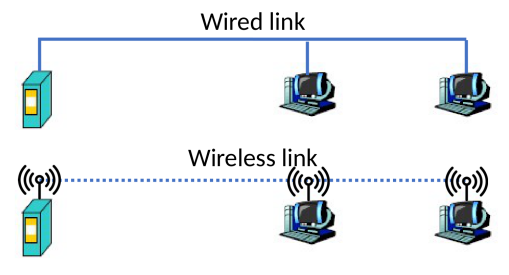
Computer traffic comes in bursts. A reserved 100 kb/s circuit on a 1 Mb/s link serves 10 users always; sharing on demand serves 35 users who are each active a tenth of the time, with the same felt performance.

Point-to-point and multipoint links

Point-to-point: a dedicated link between exactly two devices.



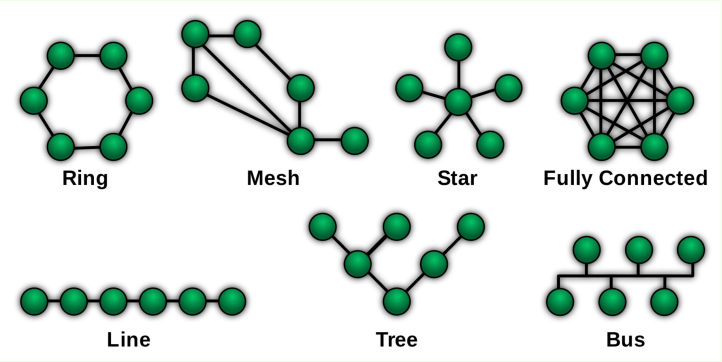
Multipoint: one link shared by several devices, wired or wireless.



If two devices on a shared link transmit at once, what happens?

Network topologies

The topology is the arrangement of the nodes and links of a network: who is physically connected to whom, independently of what runs on top.

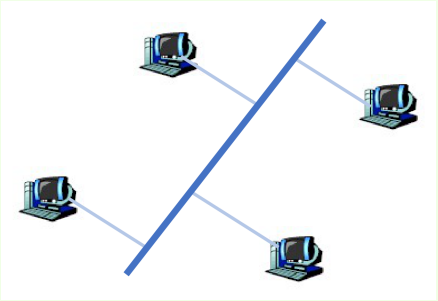


Bus topology

Every host taps into one shared backbone cable, so every transmission reaches everybody — and collisions are possible.

- Simple and cheap, good for small networks.
- A break in the bus splits the network in two.

One backbone and n drop links.

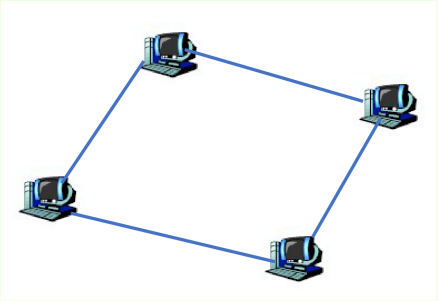


Ring topology

Each host is point-to-point connected to exactly two others, and the signal is forwarded around the ring until it reaches its destination.

- Cheap, and performs better than the bus.
- Adding a node means opening the ring.
- One misbehaving node can impair everybody.

n duplex links.

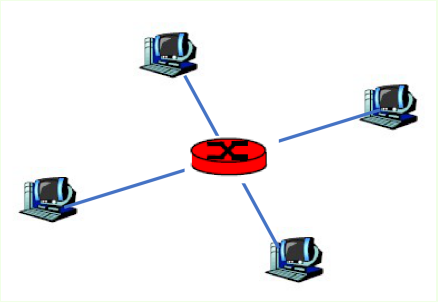


Star topology

Hosts are linked only to a central controller — a hub, a switch or a router — which redirects every message.

- Cheap, simple, scalable: adding a host adds one link.
- Every host must be able to reach the controller.
- The controller is a single point of failure.

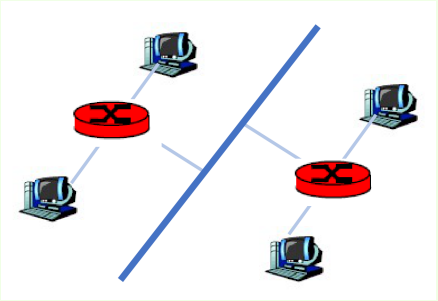
One controller and n duplex links.



Tree topology

Several stars joined through a backbone: the shape almost every real building network has.

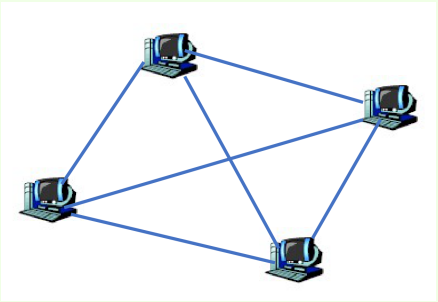
- Versatile, scalable, well supported by hardware vendors.
- Harder to configure than a single star.
- It inherits the weakness of the backbone it hangs from.



Mesh topology

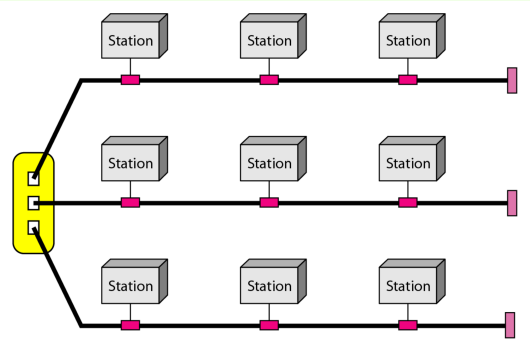
Hosts are connected point-to-point to one another, with no hierarchy: full mesh connects all pairs, partial mesh only some.

- Dedicated links, so little contention and good robustness.
- Every node needs as many ports as it has neighbours.
- It does not scale: a full mesh has $n(n-1)/2$ links.



Hybrid topologies

Real networks mix topologies: here a star backbone joining three bus networks. Each part is chosen for its own segment, and the whole is none of the component shapes.



A star backbone connecting three bus networks

Comparing topologies

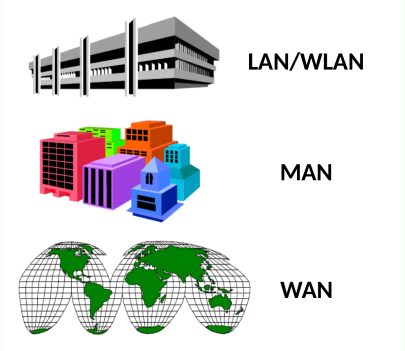
Topology	Links	If a host fails	If the medium fails
Bus	$1 + n$	no effect on the others	the network splits
Ring	n	the ring may break	the ring may break
Star	n	no effect on the others	only that host is cut off
Tree	n + backbone	no effect on the others	a whole subtree is cut off
Mesh (full)	$n(n-1)/2$	traffic takes another link	traffic takes another link

Robustness is bought with links, and links are bought with ports and cable: every real network is a compromise between these two columns.

Networks by scale

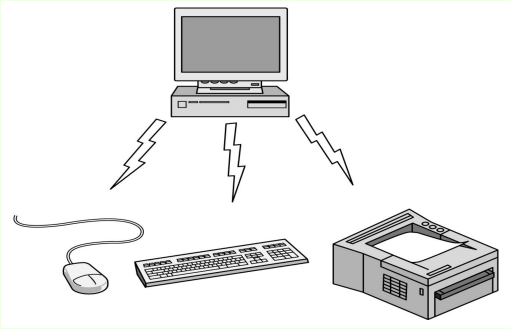
Networks are classified by the distance they span, which in turn decides the technology they can use.

- PAN — around one person, a few metres.
- LAN — one building or campus.
- MAN — a city.
- WAN — a country, a continent, or more.



Personal area networks

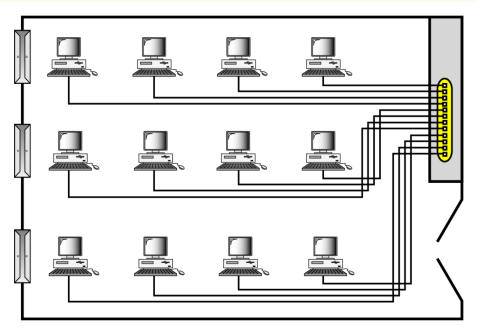
A PAN connects devices over the range of a single person: headset, keyboard, watch, printer. Bluetooth works master and slave, with the computer telling the peripherals when and on which frequency they may transmit.



A Bluetooth PAN: no cables to find and plug in

Local area networks

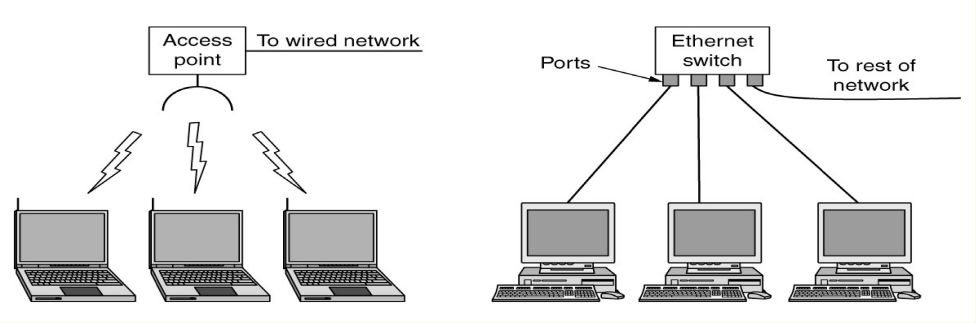
- A LAN is a private network inside one building or campus.
- Its size is bounded, so the worst-case transmission time is known in advance.
- Low latency, few errors, and today from 100 Mb/s to tens of Gb/s.



A LAN connecting twelve computers to one switch

Wireless and wired LANs

On the left an 802.11 network, where every machine talks to an access point that relays for it. On the right a switched Ethernet network, where each machine has its own link to a switch that forwards by address.



Home networks

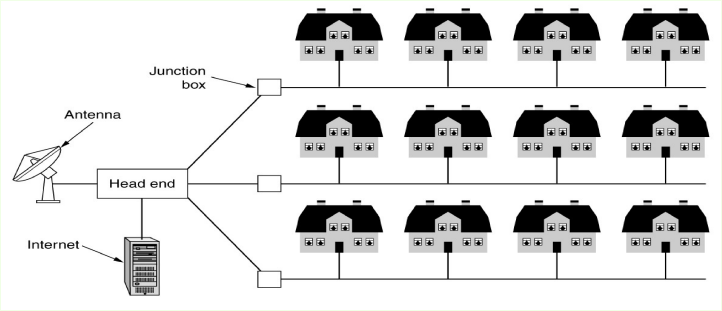
A home network is a LAN, but it is the LAN with the hardest requirements.

- It is installed by someone who will not read a manual, so it must just work.
- Its devices are bought one at a time over years and must still interoperate.
- Consumer margins are thin, and security, reliability and interoperability all cost money.
- An insecure appliance is not a lost file: it is a lock somebody else can open.

This is where the Internet of Things actually lands, and why it is a security problem before it is a networking problem.

Metropolitan area networks

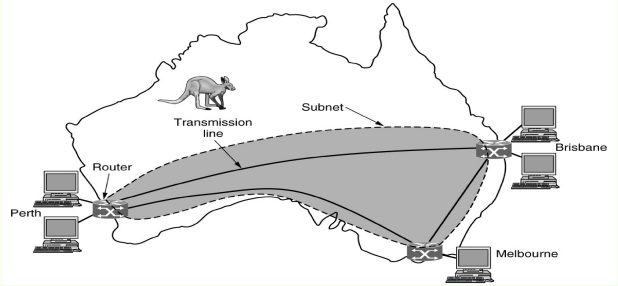
A MAN covers a city. Cable television is the best known example: built to distribute television one way, it was turned into a two-way Internet service by using the unused parts of the spectrum.



Television and Internet fed into the same cable head-end

Wide area networks: hosts and subnet

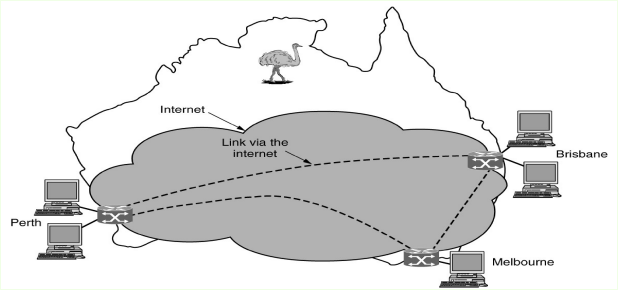
- A WAN spans a country or a continent, here three offices in Australia.
- The machines that run applications are the hosts; everything that carries their messages is the subnet.
- The subnet is transmission lines plus routers, and it is usually owned by somebody else — typically a telecommunications company.



Three offices connected by leased lines

A WAN over the public Internet

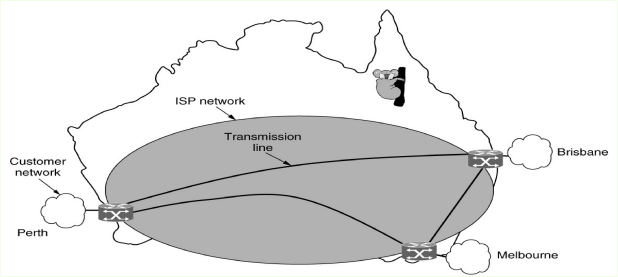
Instead of leasing lines, the offices can be joined by virtual links running over ordinary Internet connectivity. Cheaper and flexible, but the capacity is no longer yours: performance follows whatever the Internet is doing.



The same WAN built as a virtual private network

A WAN over an ISP network

A third arrangement: the customer networks connect to a provider, and the provider's own network carries the traffic between the sites and out to the rest of the Internet.



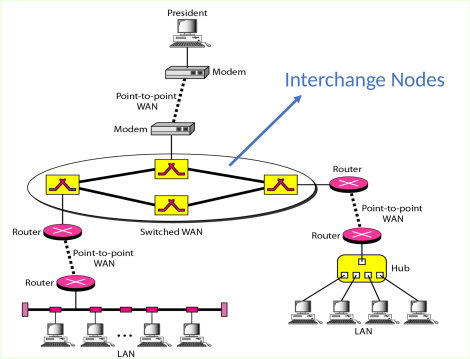
The offices connected through an ISP network

Heterogeneous WANs and internetworks

A large WAN is rarely one technology: switched cores, point-to-point links, hubs and LANs all in the same picture.

Definition

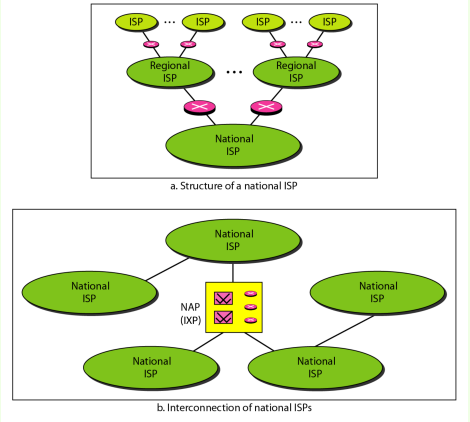
A collection of distinct, independently operated networks joined together is an internetwork. The device that joins them, and translates, is a gateway; the one that does it at the network layer is a router.



Internet Service Providers

An ISP is an organisation that provides access to the Internet: commercial, community-owned, non-profit, or a university.

- A PoP is where customers connect to the provider.
- Access, regional and national ISPs, each connecting upward.
- Providers exchange traffic at IXPs, also called NAPs.



Who connects to whom, and who pays

- Connecting every provider to every other is a mesh: unaffordable, as we saw.
- So a customer ISP pays its provider ISP, at every level of the hierarchy.
- Two providers at the same level can peer instead, connecting directly and usually paying each other nothing.
- An IXP is a building where many of them peer at once.
- An ISP can multi-home, buying from two providers and surviving one failure.

Content providers

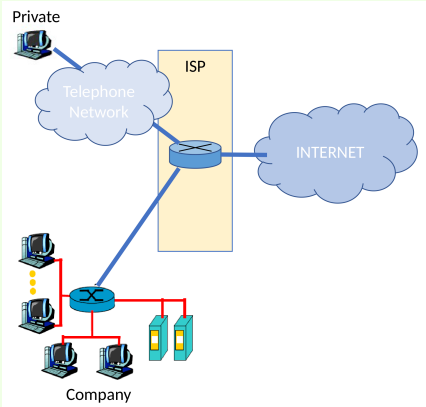
Large content providers build private global networks and peer directly with low-tier ISPs, bypassing the upper tiers: it costs them less and puts delivery under their own control.

Connecting to an ISP

Residential: a modem or an ONT reaches the provider over the existing telephone or cable infrastructure.

Business: a medium or large company buys a dedicated link straight into the ISP, and connects its own network behind a router.

Either way the first hop out of the building belongs to somebody else.



Residential access technologies

Technology	What it runs on	What to remember
DSL / ADSL	the existing telephone pair	rate falls with distance
Cable (HFC)	fibre to the street, coax in	the last stretch is shared
FTTH	fibre to the home, ONT indoors	hundreds of Mb/s, up to a gigabit
Fixed wireless	5G radio from a base station	no cable to lay

- All of them are asymmetric: downstream is faster than upstream.
- The advertised figures are a ceiling, and on a shared medium it is shared.

Summary

- A network is autonomous devices that can exchange information; hosts at the edge, packet switches in between.
- Data moves in packets, stored and forwarded, queueing when the link is busy.
- Topology says who is connected to whom; scale says PAN, LAN, MAN or WAN.
- The Internet is an internetwork: independent networks joined by routers, with money and peering deciding much of its shape.

One request crosses a dozen of these networks. How is that complexity kept manageable?

That question is the whole content of the next step: layering.